

Chi-Square Test in ML



Chi-square test in Machine learning



The **Chi-Square test** is a statistical test commonly used in **machine learning** for feature selection, particularly in categorical data. It helps to determine whether there is a significant association between two categorical variables or whether the observed frequencies deviate significantly from the expected frequencies under the assumption of independence.

Key Concepts of the Chi-Square Test

1. Hypothesis Testing:

- **Null Hypothesis (H_0):** The variables are independent (no association).
- **Alternative Hypothesis (H_1):** The variables are dependent (association exists).

2. Chi-Square Statistic Formula:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Where:

- O_i : Observed frequency.
- E_i : Expected frequency (calculated as $E_i = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$).

3. Degrees of Freedom:

- $df = (\text{rows} - 1) \times (\text{columns} - 1)$

4. P-value:

- Compares the calculated χ^2 statistic with the critical value from the Chi-Square distribution table.
- If $p \leq \alpha$ (significance level, often 0.05), reject the null hypothesis.

Application in Machine Learning

In machine learning, the Chi-Square test is often used for **feature selection** to identify features that have the strongest relationship with the target variable. This is particularly useful for classification problems with categorical data.

Steps to Perform Chi-Square Test in Feature Selection

1. Prepare the Data:

- Ensure the data has categorical features and a categorical target variable.
- Convert continuous variables into categorical ones (if needed).

2. Create a Contingency Table:

- Create a table that summarizes the relationship between a feature and the target variable.

3. Calculate Chi-Square Statistic:

- Use the formula or a library function.

4. Interpret the Results:

- Features with a higher Chi-Square statistic (and lower p-value) are more relevant.

Demo Code Using Python

python

```
import pandas as pd
from sklearn.feature_selection import chi2
from sklearn.feature_selection import SelectKBest

# Sample dataset
data = {
    'Feature1': [1, 0, 1, 1, 0, 1, 0],
    'Feature2': [0, 1, 0, 0, 1, 1, 0],
    'Target': [1, 0, 1, 1, 0, 0, 1]
}
df = pd.DataFrame(data)

# Separating features and target
X = df[['Feature1', 'Feature2']]
y = df['Target']

# Perform Chi-Square test
chi_scores, p_values = chi2(X, y)

# Display results
for i, col in enumerate(X.columns):
    print(f"Feature: {col}, Chi-Square Score: {chi_scores[i]}, P-value: {p_values[i]}")

# Feature selection using SelectKBest
selector = SelectKBest(chi2, k=1) # Select top 1 feature
X_new = selector.fit_transform(X, y)
print("\nSelected Features:\n", X.columns[selector.get_support()])
```

Output Interpretation

1. **Chi-Square Scores:** Indicate the strength of the association between each feature and the target.
2. **P-values:** Help decide statistical significance. A low p-value (e.g., < 0.05) indicates a strong relationship.
3. **Selected Features:** The most relevant features based on the test.

When to Use the Chi-Square Test?

- **Categorical Data:** When the dataset includes categorical features and a categorical target.
- **Feature Selection:** To eliminate irrelevant or less significant features.
- **Preprocessing:** Before training classification models like Naive Bayes, Decision Trees, etc.

Limitations

- Requires categorical data.
- Assumes independence between observations.
- Not suitable for continuous features without discretization.

By leveraging the Chi-Square test effectively, you can streamline your feature selection process and improve your model's performance.